A Reproducible Cross-National Dataset of Public Student Visa and Study Permit Statistics

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Abstract. In this paper, we describe the collection, cleaning, and unification of public administrative data on international student visas and study permits. The resulting artifact, `data/student_visa.sqlite`, contains 9,510,731 aggregate measure records in a 30-field long-form schema assembled from official sources for five destinations: Australia, Canada, New Zealand, the United Kingdom, and the United States. Temporal coverage extends from Australian financial year 2005–06 to 30 April 2026, Canadian records from 2015 to 2026 with outcome subsets for 2019–April 2024, New Zealand records from 2022 to 2025, UK records from 2005 to the first quarter of 2026, and US records from fiscal year 1997 to fiscal year 2024. The database retains distinct measures for applications, decisions, grants, refusals, issuances, permit holders, and reported rates instead of treating unlike official definitions as interchangeable. Origin-country labels are unified against a preserved 248-record United Nations M49 reference using reviewed aliases and explicit statuses for direct matches, alias matches, aggregates, unknown values, and unmatched labels. The preparation process is reproducible from preserved raw sources with a single build command and separates extraction, transformation, normalization, database loading, and validation. The collected data can support analysis of the implementation and evolution of student-migration policies and administrative practices across major study destinations, as well as carefully scoped comparisons among them. Source-specific differences in legal categories, applicant populations, time bases, origin concepts, and statistical denominators remain explicit in the artifact.

Keywords: Educational migration · Student visa · Open data collection · Historical data

1 Introduction

International education research often begins with enrolment, mobility, or institution-level data. A separate transition occurs after admission or eligibility to study and before study begins: an applicant requests a student visa, study permit, or equivalent authorization and receives a government decision. Public statistics describing this transition can inform migration research, policy evaluation, and international-education planning. However, no assumption can be made that two official tables bearing similar labels represent the same event or denominator.

The underlying data are fragmented institutionally and technically. Immigration New Zealand publishes annual offshore or overseas decision tables on web pages [3]. The Australian Department of Home Affairs distributes large pivot-style workbooks covering lodgements, grants, and decision-based grant rates [4]. The UK Home Office publishes broad entry-clearance and sponsored-study workbooks that require category and applicant filtering [5]. Canadian records span committee-transparency pages and open-data files with different populations and origin concepts [6]. The US Department of State publishes visa issuances by nationality and visa class, but not nationality-level refusal denominators in the same table [7].

This work treats the dataset itself as a research output. Its objective is to determine what reusable, origin-aware student-visa data can be reconstructed from these public government sources while preserving their statistical meanings. This approach is consistent with the broader principle that reusable data should be findable, accessible, interoperable, and reusable, while recognizing that openness alone does not create semantic comparability [1].

The contributions are:

- a preserved inventory of official source files and their source-level schemas;
- reproducible extraction of HTML, CSV, ordinary XLSX, and XLSX pivot-cache records;
- a 30-field long schema that retains source-specific dimensions and measure definitions;
- destination-aware origin normalization against a preserved UN M49 reference [2];
- a strict SQLite artifact that excludes non-study proxies and explicitly identifiable dependant cohorts; and
- executable validation and provenance audits rather than undocumented manual cleaning.

The dataset is intended for aggregate analysis. It contains no personal records, does not represent individual application histories, and must not be used as individualized visa advice or as evidence of discrimination or causality without a separately designed study.

The code and documentation are maintained in the *study-visa-open-data* GitHub project [8]. Unless an absolute location is explicitly given, every relative file and directory path in this article is interpreted from the root of that project.

2 Data Practices in Related Research

3 Data Sources

3.1 Data source references

We used official web pages and downloadable resources published by national immigration authorities and other government organizations as the sources of data. At the time of collection, all materials were publicly available without registration, authentication, or special permission. The retrieval and extraction scripts provided in the project repository accessed and processed the sources in their published form and operated successfully at the time of data collection.

Table 1 summarizes the data sources. The complete inventory, including URLs, local filenames, formats, fields, reporting periods, and comparability notes, is machine-readable in `data/country-schema.json`.

3.2 Source acquisition and provenance

The separate data artifact contains the raw snapshots used for this version; its DOI will be added when available. Retrieval commands and official URLs are documented in `data/RETRIEVAL.md`; source metadata and schemas are recorded in `data/country-schema.json`. Thus, the end-to-end build reproduces derived files from preserved artifact inputs, or from sources retrieved using the documented commands. It does not claim that a future rerun of every remote URL will retrieve identical bytes, because government resources and dashboard endpoints may change.

4 Collection Methodology

Collection was designed around preserved public records and a machine-readable source inventory. Official government sources were located by publisher, statistical topic, reporting period, and availability of downloadable tables. Each retained source is identified by its official URL, publisher, access or snapshot date, local filename, format, reporting coverage, and source-specific definitions in `data/country-schema.json`. Retrieval commands are recorded in `data/RETRIEVAL.md`. Where a visa-statistics source was distributed as a static CSV, XLSX, or HTML file, it was downloaded and preserved unchanged; transformations operate on repository copies rather than altering the raw input. The present release was assembled with Python 3 from source snapshots available by 19 July 2026.

4.1 Executable build chain

The supported build is launched from the repository root:

```
python3 scripts/rebuild_database.py
```

The driver stops if any child process fails. Table 2 summarizes the seven documented stages. The same information, including initial parameters, inputs, outputs, and observed row counts, is available in JSON form at `scripts/collect_data_script_chain.json`.

For a faster rebuild that reuses the source-level CSV extractions, the driver accepts the `--skip-extraction` option. Validation can be disabled with `--skip-validation`, although this is not recommended for a release build.

Table 1. Official data sources and their role in the dataset.

Destination	Publisher	Retrieved data	Principal qualification
New Zealand	Immigration New Zealand [3]	Annual offshore/overseas applications, approvals, declines, and rates, 2022–2025	The 2022 schema differs from 2023–2025; coverage is offshore/overseas only.
Australia	Department of Home Affairs [4]	Lodgement, grant, and grant-rate XLSX pivot workbooks, FY2005–06 through 30 April 2026	Grant rate is grants divided by grants plus refusals, not by lodgements. Only primary applicants enter generated analytical artifacts.
United Kingdom	Home Office [5]	Entry-clearance applications/outcomes and sponsored-study course-level workbooks, 2005–2026 Q1	Broad entry-clearance data require a Study-category filter. Only main applicants are retained; course-level tables cover CAS-matched main applicants.
Canada	Immigration, Refugees and Citizenship Canada [6]	Approval/outcome snapshots, intake and issuance tables, permit-holder files, and a French/bilingual subset, 2015–2026 with outcome subsets for 2019–April 2024	Country of residence and citizenship are different origin concepts. Recent outcome tables have restricted populations or top-country coverage.
United States	Department of State [7]	F1 and M1 issuances by nationality and fiscal year, FY1997–FY2024	Issuances do not provide nationality-level application/refusal denominators; F2 and M2 dependants are excluded.

Table 2. Reproducible preparation chain.

Step	Script	Function
1	`extract_csv.py`	Extract New Zealand, Australian, UK, and Canadian source tables.
2	`extract_us.py`	Extract United States F1 and M1 issuances.
3	`combine_canada.py`	Combine Canadian outcome, intake/issuance, and permit-volume source tables without conflating their coverage.
4	`extract_canada_recent.py`	Produce the Q-2700 French-speaking/bilingual subset for SQLite-only loading.
5	`build_common_table.py`	Transform supported records to the common long schema and apply applicant-scope rules.
6	`load_sqlite.py`	Apply the strict category filter, load a temporary database, create indexes, and atomically replace the final database.
7	`validate_database.py`	Validate references, measures, mappings, provenance, reconciliation rules, and applicant scope; write the unmatched-label audit.

4.2 Format-specific extraction

HTML tables are parsed from the preserved government pages. Ordinary CSV and XLSX tables are read according to their documented sheet and field layouts. Australian workbooks require a different method: the visible sheets are pivot reports rather than flat source tables. The extractor reads the underlying XLSX pivot-cache definitions and records directly. This recovers dimensions that would be lost by copying visible cells, including lodgement channel and, for the grants workbook, the last visa held.

The extracted Australian source tables contain 2,079,645 lodgement records, 2,274,440 grant records, and 2,046,442 grant-rate/decision records before primary-applicant filtering. The UK extraction produces 249,920 broad entry-clearance application records, 402,372 broad outcome records, 18,717 sponsored-study course application records, and 18,658 course-level grant records before strict category and applicant filtering. These source-level counts should not be interpreted as numbers of individual applicants: each record is a dimensional aggregate and later may emit one or more long-form measures.

US workbook cells representing counts are normalized to displayed integer values. Missing or suppressed values remain missing rather than being converted to zero. Region totals and grand totals are excluded, as are dependent visa classes F2 and M2.

5 Data Processing

Processing converts heterogeneous extracted tables into scoped analytical records through explicit applicant and visa-category filters, country-name reconciliation, long-form transformation, and atomic loading. Incompatible measures remain separate, and unavailable values are not imputed.

5.1 Applicant and category scope

The strict analytical artifacts exclude explicitly identifiable dependant or mixed-applicant cohorts:

- Australian source extractions preserve both values for provenance, but common outputs retain `Applicant Type = Primary`;
- UK entry-clearance source extractions preserve `All`, `Main Applicant`, and `Dependant`, but common outputs retain `Main Applicant`; sponsored-study course tables are already main-applicant-only; and
- US outputs retain the primary student classes F1 and M1 and exclude F2 and M2.

Sources without a separable applicant-type field remain as published. Consequently, absence of an explicit dependant label is not proof that every aggregate from New Zealand or Canada represents primary applicants only.

The database loader additionally requires the local student/study category for each destination. It excludes non-study UK entry-clearance categories. The Canadian Q-2700 language subset is loaded only into SQLite and retains a distinct coverage label so that it is not silently merged with all-applicant Canadian outcomes.

5.2 Source differences and structural missingness

The common schema is the union of dimensions and measures exposed by five national statistical systems, not a claim that every source supplies every field. An empty dimension therefore usually means that the attribute was not published or was not applicable to that source. If a measure is unavailable, the long-form transformation emits no row for that measure rather than creating a row with a blank value or replacing it with zero. Derived values are produced only where the available components support the official definition.

New Zealand. The annual Immigration New Zealand tables describe offshore or overseas decisions by origin and calendar year, so fields for financial year, quarter, month, education sector or level, age, gender, provider state, and lodgement channel remain empty. The 2023–2025 tables report completed-application volume, approvals, declines, approval rate, and decline rate, but the 2022 table reports only approvals, declines, and decline rate. A 2022 completed-application total could be calculated as approvals plus declines, and an approval rate could then be calculated from that total, only if those two outcomes exhaust the source population. Because the source does not establish that withdrawals or other outcomes are absent, the present build does not make that assumption: 2022 application-volume and approval-rate rows remain unavailable.

Australia. The Department of Home Affairs publishes separate monthly pivot workbooks for lodgements, grants, and decision-based grant rates. Their rich dimensions populate financial year, quarter, month, client location, lodgement channel, sector, applicant type, provider state, gender, age, and citizenship, but they do not provide a separate education-level field; the grants workbook alone exposes last visa held. Lodgements, grants, and decisions are different events and are therefore not used to fill one another's missing measures. In the decision workbook, the source provides grants, refusals, and total decisions; validation confirms that grants plus refusals equals that total, and the pipeline calculates grant rate as grants divided by grants plus refusals. The dedicated grants workbook and the grant component of the decision workbook remain distinct because their dimensional structures differ.

United Kingdom. UK data are quarterly and nationality-based. General entry-clearance files provide applications and decision outcomes, while sponsored-study files provide course-level applications and grants only from 2018 onward. Consequently, education level is populated for the course-level series but empty for general entry-clearance records, and fields such as applicant location, lodgement channel, age, gender, and provider state are not available. Applications and outcomes published in the same quarter do not necessarily describe the same application cohort, so the pipeline does not derive a quarterly approval rate by dividing grants by applications. Refused, withdrawn, and lapsed decisions remain separate published outcome measures.

Canada. Canadian coverage is assembled from several sources with different populations and origin concepts. Older outcome tables use country of residence and may report approvals, refusals, withdrawals, processed totals, and approval rates; recent snapshots may cover only leading countries, aggregates, or French-speaking/bilingual applicants. Separate Open Canada files report permit holders or permits effective by citizenship rather than applications and decisions. These differences leave many dimensions and measures empty for particular Canadian rows and prevent one source from completing another. Source-reported approval rates are retained with their stated denominator. In the Q-2700 French-speaking/bilingual subset, processed applications can be derived as approvals plus refusals because those components define the reported decision total, but that subset remains explicitly labelled and SQLite-only rather than being treated as an all-applicant national series.

United States. The Department of State workbooks provide annual F1 and M1 visa issuances by nationality and fiscal year. They do not provide nationality-level applications, refusals, withdrawals, decisions, or approval rates in the same source, nor do they expose education sector, level, applicant location, age, gender, provider state, or lodgement channel. Those fields therefore remain empty or have no corresponding measure rows. An approval rate cannot be responsibly derived from issuances alone because its application or decision denominator is unavailable.

5.3 Country-name reconciliation

Origin labels are normalized against the United Nations Statistics Division M49 methodology table [2]. The official source was downloaded on 19 July 2026 as `data/UNSDâĂŤMethodology.csv` from the UNSD M49 methodology page [2] and is used directly by the normalization scripts. The table contains 248 unique countries or areas and records the canonical name, three-digit M49 code, region, subregion, intermediate region, and ISO alpha-2 and alpha-3 codes.

Normalization first applies destination-specific or global reviewed rules from `data/country_aliases.csv`, then exact and normalized-form matching against the M49 canonical names. Each source label receives one of five statuses:

matched
: an official M49 name matches directly or after orthographic normalization;

alias_matched
: a reviewed source-specific alias maps to an M49 name;

aggregate
: the label denotes a total, region, or other non-country group;

unknown
: the source explicitly denotes unknown or unstated origin; or

unmatched
: no responsible canonical assignment is available.

Table 3 reports the resulting distribution in the 9,510,731-row strict SQLite database. These are measure-row counts rather than counts of distinct source labels or applicants, so Australia's highly dimensional records have a large influence on the percentages.

Table 3. Origin-country normalization results in the strict SQLite database.

Normalization status	Rows	Share (%)	Canonical `un_m49_country`
`matched`	6,806,596	71.568	Assigned from the M49 reference
`alias_matched`	2,416,032	25.403	Assigned through a reviewed alias
`aggregate`	1,459	0.015	Not assigned; label is not a country or area
`unknown`	20,542	0.216	Not assigned; source reports unknown origin
`unmatched`	266,102	2.798	Not assigned; no responsible mapping available
Total	9,510,731	100.000	—

Only `matched` and `alias_matched` rows receive `un_m49_country`. Territories represented separately by UNSD are not reassigned to an administering state. The 266,102 unmatched rows collapse to 59 destination/source/label groups and are exported with destination, original label, source, row count, and period range to `log/unmatched_origin_labels.csv`.

5.4 Long-form transformation

The original government files organize their statistics in different ways. Some place several measures in one record. For example, one Australian decision record can contain grants, refusals, total decisions, and grant rate. The transformation converts that source record into four long-form rows: one row for each measure. Every emitted row retains the same source dimensions, such as origin country, period, education sector, applicant location, gender, and age group. This structure allows different national measures to use one common schema without implying that measures with similar names have identical definitions.

Rows must nevertheless be aggregated within the correct source and coverage. Australian grants, for example, appear in both the dedicated grants workbook and the decision/grant-rate workbook. The two representations have different dimensional structures and are retained for different analytical purposes. Summing them together would count the same type of event twice. Queries must therefore select the intended source, measure, period, applicant scope, and relevant dimensions before calculating totals or rates.

6 Dataset Description

6.1 Artifacts

The release is organized in layers (Table 4). Raw files and extracted source tables remain distinguishable from harmonized records.

6.2 Database coverage

Table 5 reports database counts obtained from the rebuilt SQLite artifact. A "record" is a measure row, not a unique person, application, or decision. Distinct origin labels include aggregates and historical labels; distinct M49 values count normalized countries or areas only.

Australia accounts for 98.0% of database records. This reflects the dimensional granularity of its pivot caches and the emission of multiple measures from decision rows, not a statement that 98.0% of international student visa applicants apply to Australia. Analyses should group or sample at an explicitly defined statistical level rather than treating every long-form row as exchangeable.

6.3 Common schema

All SQLite columns are stored as text to preserve source formatting and allow uniform bulk loading. Consumers should cast `measure_value` according to `measure_unit`; validation ensures counts are non-negative integer strings and percentages lie between 0 and 100. The 30 fields are grouped in Table 6.

Table 4. Principal repository artifacts.

Artifact	Records	Scope
data/original/<country>/	—	Preserved official raw files; heterogeneous formats and schemas.
data/csv/<country>/	source-specific	Extracted source-level tables, including broad categories retained for provenance.
data/student_visa_common_long.csv	9,707,903	Thirty-field common long table for five destinations; retains distinguishable source categories and excludes identifiable dependant cohorts.
data/student_visa.sqlite	9,510,731	Strict student/study query database for five destinations; includes the labelled Canadian Q-2700 SQLite-only subset.
log/unmatched_origin_labels.csv	59 groups	Audit of source-label/source combinations not assigned an M49 country.

Table 5. Coverage of the strict SQLite database.

Destination	Records	Origins	Period	Included measures
Australia	9,319,202	236/224	FY2005–06 to 30 Apr 2026	Lodgements, grants, decisions, refusals, and grant rates
Canada	54,729	283/235	2015–2026; outcome subsets 2019–Apr 2024	Applications, outcomes, issuances, and permit-holder/effective counts
New Zealand	2,777	180/178	2022–2025	Applications, approvals, declines, and rates
United Kingdom	122,895	212/203	2005–2026 Q1	Study applications/outcomes and CAS-matched course grants
United States	11,128	211/195	FY1997–FY2024	F1 and M1 issuances
Total	9,510,731	—	—	—

Table 6: Common long-table data dictionary.

Group	Fields	Interpretation
Identity and origin	`destination_country`, `origin_country`, `un_m49_country`, `normalization_status`, `origin_country_type`	Destination; source label; normalized M49 name; mapping status; and whether origin means citizenship, nationality, residence, or another published concept.
Time	`period_label`, `period_type`, `calendar_year`, `financial_year`, `quarter`, `month`	Original period representation plus parsed components where meaningful. Blank fields are explicit non-applicability or unavailable detail.
Visa and applicant	`visa_category`, `visa_category_source`, `applicant_location`, `lodgement_channel`, `applicant_type`	Harmonized and original legal/category labels, location, application channel, and applicant population.
Education and demographics	`education_sector`, `education_level`, `gender`, `age_group`, `provider_state`, `last_visa_held`	Optional dimensions retained when a source exposes them; not uniformly available across destinations.
Measure	`measure_type`, `measure_value`, `measure_unit`, `rate_denominator_type`	Published or computed statistic, value, count/percent unit, and explicit rate denominator when known.
Coverage and provenance	`coverage_type`, `source_snapshot`, `source_file`, `source_notes`	Population/subset label, source release date, repository-relative lineage, and qualifications.

6.4 Indexes and query access

SQLite indexes support destination, raw origin, origin type, period, measure, visa category, normalized M49 country, normalization status, and the composite destination–origin–measure path. A typical scoped query is:

```
SELECT financial_year, origin_country, measure_type, measure_value
FROM student_visa_common_long
WHERE destination_country = 'Australia'
  AND applicant_type = 'Primary'
  AND education_sector = 'Higher Education Sector'
  AND measure_type IN ('decisions', 'grants', 'refusals');
```

The query intentionally selects the dimensional and measure fields required for interpretation. Summing `measure_value` without constraining source, measure, period, and relevant dimensions may double count published aggregates.

6.5 Coverage and update frequency

Coverage is historical but source-dependent: the earliest included US fiscal year is 1997, while other destination series begin later and may end in a quarter, month, partial year, or source-specific snapshot. There is no single update frequency for the unified dataset. Australian program workbooks are released periodically with monthly cut-offs; UK and New Zealand tables follow their official statistical releases; and several Canadian components are annual, periodic, or one-off transparency extracts. The database is therefore versioned as a snapshot and updated by rerunning the complete build against newly preserved source releases, not by assuming a universal calendar schedule.

7 Technical Validation

Validation is part of the default build rather than a post hoc manual review. Verification covers reference-schema integrity, dimensional-key uniqueness, decision reconciliation, numeric domains, provenance completeness, normalization consistency across artifacts, and exclusion of explicitly identifiable dependant cohorts. The release passed the checks detailed below.

7.1 Reference and normalization checks

The retained M49 table must contain the required schema, 248 unique non-empty country or area names, and valid three-digit M49 codes. Alias tests verify reviewed Australian label mappings, distinguish non-country labels, and ensure separately listed territories remain distinct. Normalization results must agree across the common CSV, the Canadian SQLite-only subset, and SQLite. Exact official M49 names may not remain unmatched, and aggregate, unknown, and unmatched labels may not receive a canonical country.

The normalization distribution is reported in Table 3. The 59 unmatched destination/source/label groups chiefly involve labels absent from the supplied current M49 list, including historical countries or areas, subnational areas, Taiwan, and Kosovo; the pipeline leaves them explicit rather than silently imposing a contested or historically incorrect mapping.

7.2 Source-structure and arithmetic checks

For the audited Australian Higher Education/Primary slice, records must be unique on all pivot-cache dimensions. This check guards against the earlier omission of lodgement channel and last-visa-held fields. Every Australian grant-rate record must satisfy

$$\text{grants} + \text{refusals} = \text{decisions}. \tag{1}$$

Count values in the Australian source tables and the final database must be non-negative integers. Percentage values must lie in the closed interval $[0, 100]$. Every SQLite row must contain a repository-relative source-file value.

7.3 Scope checks

The common CSV and SQLite are scanned for Australian `Secondary`, UK `Dependant` and `All`, generic `dependent`, and US F2/M2 records. The validated count of such explicitly excluded records in SQLite is zero. The rebuilt SQLite table contains 9,510,731 rows after strict filtering and insertion of the explicitly labelled Canadian SQLite-only subset.

Four unit tests cover reviewed aliases, non-country handling, territory preservation, and equality between generated and source M49 names/counts. The full build and all validation checks completed successfully for the 19 July 2026 snapshot.

8 Usage Notes

8.1 No universal approval-rate denominator

The dataset deliberately avoids constructing a universal `approval_rate`. New Zealand's recent tables relate completed applications to approvals and declines. Australia's grant-rate workbook defines the denominator as grants plus refusals. UK applications and decisions can occur in different quarters. Canadian tables include processed outcomes, permit issuances, permit holders, and subsets with different coverage. The US table contains issuances without nationality-level application or refusal totals. These measures must not be substituted for one another.

8.2 Origins are not semantically identical

Depending on the source, origin is citizenship, nationality, country of residence, or an aggregate region. For Canada in particular, approval/outcome tables use country of residence while permit-holder files use citizenship. The `origin_country_type` field must be included in joins and comparisons. A common M49 label reconciles spelling; it does not make residence equivalent to citizenship.

8.3 Populations and coverage vary

New Zealand data cover offshore or overseas applications. Australia is limited in generated artifacts to primary applicants, but its three workbook families have different event definitions and dimensions. UK course-level records include main applicants matched to Confirmation of Acceptance for Studies data and should not be treated as a complete copy of entry-clearance totals. Recent Canadian tables may cover top countries, aggregates, or French-speaking/bilingual applicants rather than all applicants. US records cover F1/M1 visas issued, not all study authorizations or decisions. Applicant-type fields are absent from some sources, so primary-only coverage cannot always be verified.

8.4 Time bases and policy changes

The collection mixes calendar years, calendar quarters, fiscal years, financial-year months, partial years, and source-defined reporting periods. Visa categories and policy rules may change within the historical coverage. Researchers should use the parsed time fields together with `period_label`, `source_snapshot`, and source notes, and should not assume that a category name is legally stable over two decades.

8.5 Long-form duplication and weighting

Rows are published measures at dimensional keys, not statistically independent observations. One source record can emit several measure rows, and similar concepts can appear in separate official workbooks. Australia dominates the row count because it exposes more dimensions. Model training, descriptive summaries, and visualizations therefore require a defined unit of analysis and explicit aggregation rules; raw row weighting would mostly measure publication structure.

8.6 Missingness, suppression, and normalization

Blank fields are not imputed. A blank can mean unavailable, suppressed, not applicable, or absent from a source schema; the source documentation should be consulted before assigning a missing-data mechanism. Aggregate, unknown, and unmatched origin statuses remain distinct. Users may propose additional aliases, but changes should be reviewed and recorded rather than implemented through ad hoc string replacement.

8.7 Appropriate and inappropriate uses

Appropriate uses include audits of public-data availability, destination-specific descriptive statistics, reproducibility studies, and comparisons restricted to demonstrably compatible measures and populations. The resource is not suitable by itself for individualized probability estimates, causal attribution, or claims about discriminatory treatment. Such research would require careful population definitions, exposure denominators, policy controls, and an ethical and legal design beyond this data descriptor.

9 Data and code availability

The primary release file is `data/student_visa.sqlite`. The database, common CSV, raw sources, source-level CSVs, and validation audit are distributed as a separate data artifact (DOI forthcoming), rather than committed to the GitHub code repository [8]. The code repository contains the scripts, machine-readable source inventory at `data/country-schema.json`, retrieval documentation at `data/RETRIEVAL.md`, and executable workflow description at `scripts/collect_data_script_chain.json`. The data files can be obtained from the artifact or recreated with the documented retrieval and build scripts.

At the time of this draft, no archival DOI or dataset version identifier has been assigned. These items should be added before submission or public release. Licences and attribution requirements of each underlying government source remain applicable independently of the project-wide reuse licence assigned to the repository or any licence later assigned to the derived database.

10 Reproducibility checklist

A release can be independently checked from the repository root as follows:

```
python3 -m json.tool data/country-schema.json
python3 -m json.tool scripts/collect_data_script_chain.json
python3 scripts/rebuild_database.py
python3 -m unittest discover -s tests -v
sqlite3 -readonly data/student_visa.sqlite \
  "SELECT COUNT(*) FROM student_visa_common_long;"
```

The expected final query result for this snapshot is `9510731`. Reproduction requires Python 3, the preserved files under `data/original/` obtained from the separate artifact or retrieved using `data/RETRIEVAL.md`, and the official reference at `data/UNSDâĂŤMethodology.csv`. The documented build scripts use the Python standard library.

11 Conclusion

This dataset demonstrates that a substantial origin-aware resource can be reconstructed from public student-visa and study-permit statistics, but also that nominally similar government measures are not automatically comparable. The central methodological result is therefore both an artifact and a boundary: 9.5 million strictly scoped measure records can be queried through a common schema, while source-specific legal categories, populations, origin concepts, time bases, and denominators remain visible. Preserving those distinctions makes the resource more useful for responsible comparative research than a smaller but falsely uniform approval-rate table.

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